MITRE ATLAS | AML.CS0003 - Bypassing Cylance's AI Malware Detection

Entry Information

ID:
AML.CS0003

Type:
Case-Study

Name:
Bypassing Cylance's AI Malware Detection

URL:
https://atlas.mitre.org/studies/AML.CS0003/

Incident Date:
2019-09-07

Target:
CylancePROTECT, Cylance Smart Antivirus

Actor:
Skylight Cyber

Description

Researchers at Skylight were able to create a universal bypass string that evades detection by Cylance's AI Malware detector when appended to a malicious file.

Attack Procedure:
[AML.T0000] Search Open Technical Databases: The researchers read publicly available information about Cylance's AI Malware detector. They gathered this information from various sources such as public talks as well as patent submissions by Cylance.
[AML.T0047] AI-Enabled Product or Service: The researchers had access to Cylance's AI-enabled malware detection software.
[AML.T0063] Discover AI Model Outputs: The researchers enabled verbose logging, which exposes the inner workings of the ML model, specifically around reputation scoring and model ensembling.
[AML.T0017.000] Adversarial AI Attacks: The researchers used the reputation scoring information to reverse engineer which attributes provided what level of positive or negative reputation.
Along the way, they discovered a secondary model which was an override for the first model.
Positive assessments from the second model overrode the decision of the core ML model.
[AML.T0043.003] Manual Modification: Using this knowledge, the researchers fused attributes of known good files with malware to manually create adversarial malware.
[AML.T0015] Evade AI Model: Due to the secondary model overriding the primary, the researchers were effectively able to bypass the ML model.

References

- Skylight Cyber Blog Post, "Cylance, I Kill You!" — https://skylightcyber.com/2019/07/18/cylance-i-kill-you/

- Statement's from Skylight Cyber CEO — https://www.security7.net/news/the-new-cylance-vulnerability-what-you-need-to-know